



Cooperation

IB 431 – Behavioral Ecology
Fall 2025
September 30th 2025

Outline

- I. Examples of cooperation within and across species
- II. Evolution of cooperation
- III. Discuss third proposal assignment (Hypotheses and predictions)
- IV. Discuss Bshary et al 2006

Examples of cooperation

- Within species
 - **Hunting in groups – dolphins, wolves, etc.**



Examples of cooperation

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 - **Vampire bats sharing bloodmeals**



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 - Hunting in groups – dolphins, wolves, etc.
 - Vampire bats sharing bloodmeals
 - **Division of labor in social insects**



Examples of cooperation

- Within species
 - Hunting in groups – dolphins, wolves, etc.
 - Vampire bats sharing bloodmeals
 - Division of labor in social insects
 - **Alarm calls in meerkats (sentinels)**



Examples of cooperation

- Across different species
 - **Cleaner wrasse and larger fish**



Examples of cooperation

- Across different species
 - Cleaner wrasse and larger fish
 - **Plovers and crocodiles**



Examples of cooperation

- Across different species
 - Cleaner wrasse and larger fish
 - Plovers and crocodiles
 - **Mixed-species schools or flocks**



Examples of cooperation

- Across different species
 - Cleaner wrasse and larger fish
 - Plovers and crocodiles
 - Mixed-species schools or flocks
 - **Prawn goby and pistol shrimp**



Why should animals cooperate?

- Meerkats that send alarm calls are at higher predation risk – how could this evolve?



Kin selection

- Meerkats typically live in closely related groups
 - Example of Hamilton's rule

Cooperation is favored when: **$rB - C > 0$**

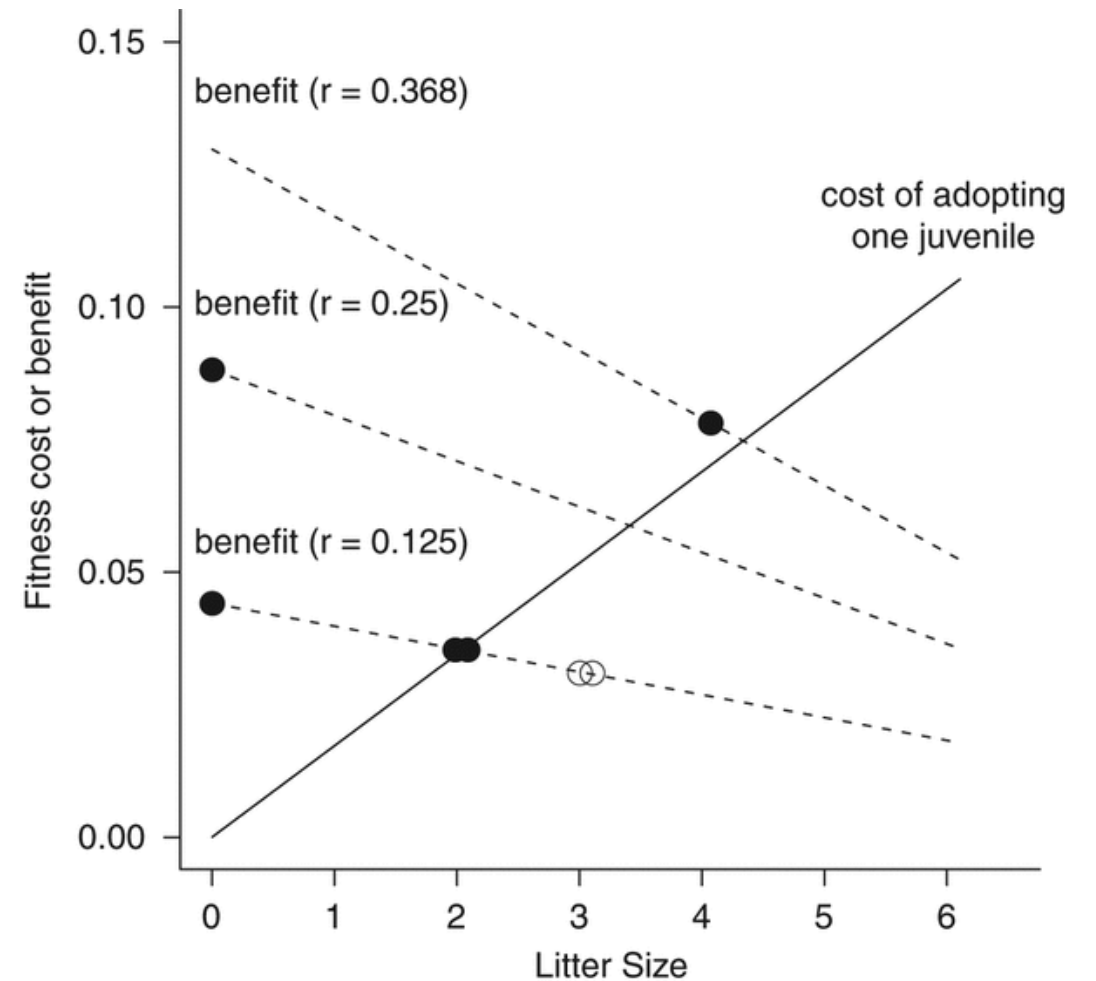
r = relatedness between altruist and recipient

B = fitness benefit to recipient

C = fitness cost to the altruist

Kin selection

- Squirrels will adopt orphaned or abandoned juveniles if they are closely related



Gorrell et al 2010

Kin selection in cooperatively breeding species

- Offspring may forgo their own reproduction to help parents and/or siblings reproduce

Kin selection in cooperatively breeding species

- Subordinate males may help dominant males attract mates
 - Wild turkeys
 - Manakins



Krakauer 2005



Kin selection in cooperatively breeding species

- Kingfishers
 - Primary helper: stay to help parents
 - Secondary helper: help an unrelated mating pair



Reyer 1984

Kin selection in cooperatively breeding species

- Kingfishers
 - Primary helper: stay to help parents
 - Secondary helper: help an unrelated mating pair
 - **What might be the benefit of helping unrelated individuals?**



Reyer 1984

Why should *unrelated* animals cooperate?

- Inheritance of resources
- Gain experience to later move up to dominant role



Reyer 1984

Why should *unrelated* animals cooperate?

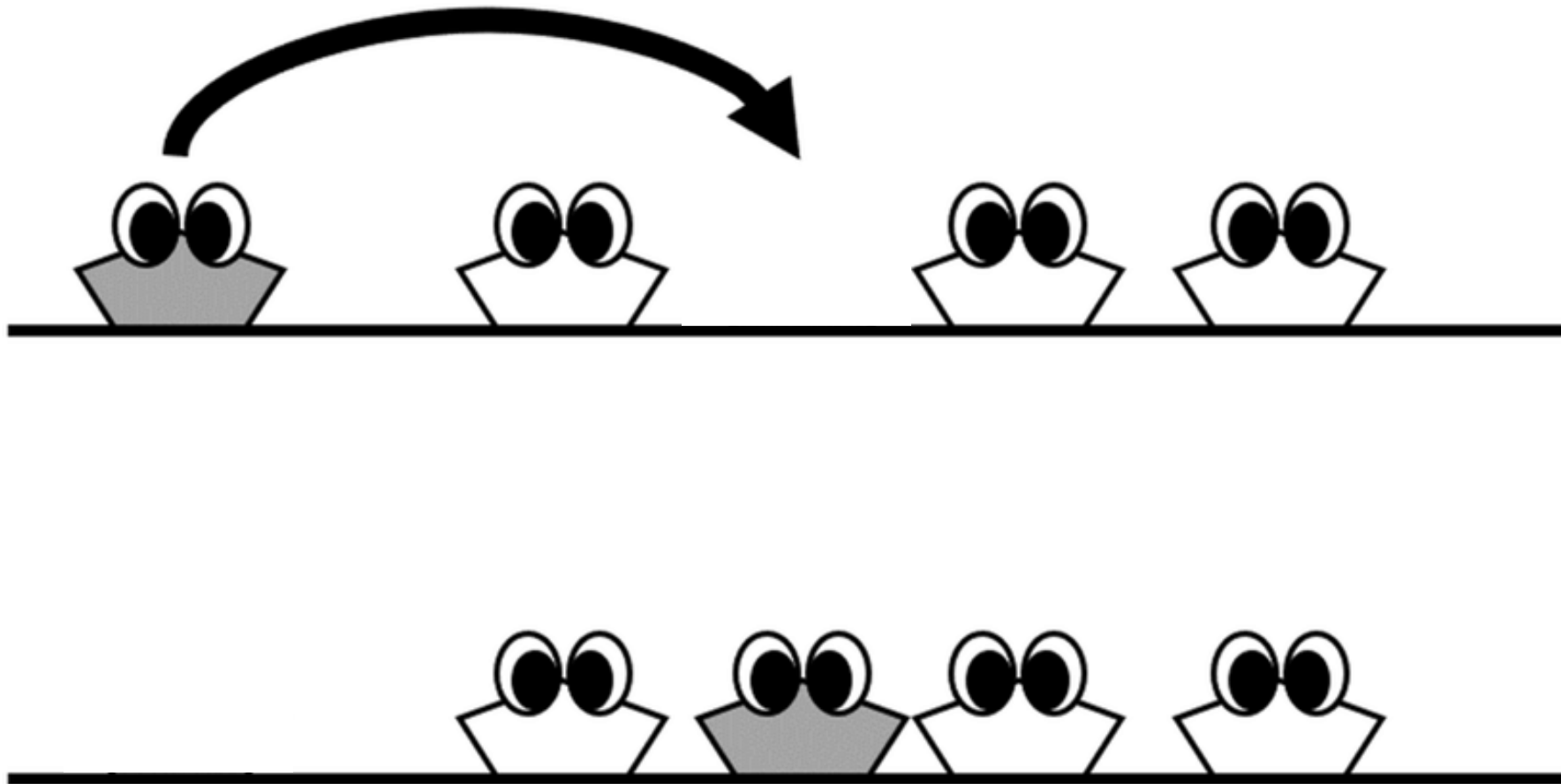
- Inheritance of resources
- Gain experience to later move up to dominant role
- **Cooperation may be byproduct of selfish behavior**

Selfish herd hypothesis

- Animals across many taxa form groups to reduce predation risk
 - Are herds of gazelle or schools of fish working together to help one another avoid predators?



Selfish herd hypothesis



Cooperation as byproduct of selfish behavior

- Demand for the same resources brings groups together
 - Reptiles basking in sunny areas
 - Zebras drinking from a swimming hole



Why should *unrelated* animals cooperate?

- Inheritance of resources
- Gain experience to later move up to dominant role
- Cooperation may be byproduct of selfish behavior
- **Social consequences for cheating / non-cooperators**

Consequences for non-cooperators

- **Tit-for-tat**
 - Reduced interaction or cooperation with individuals that did not previously cooperate
 - Requires individual recognition



Tit-for-tat in baboons

- 250,000 tests of random pairings over 95 days
 - Solo reward or both get reward
 - Partner gets reward or nothing
 - Non-cooperators were less likely to get reward from partner or had partner leave



Formaux et al 2023

Consequences for non-cooperators

- Policing in ants
 - Obligate vs. facultative eusociality
 - Workers sometimes able to reproduce but suppressed via hormones produced by queen



Smith et al 2008

Consequences for non-cooperators

- *Aphenogaster cockerelli* ants
 - Applied scent present only in fertile queens to workers
 - These workers were were bitten, pulled and held by other workers
 - Unless there was no queen present



Smith et al 2008

Game theory and cooperation

- Game theory models (prisoner's dilemma, tit for tat, hawk-dove, etc.) can explain how cooperation might emerge
- Coming soon – guest lecture on modeling, game theory, and cooperation by Dr. Tina Barbasch

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Hypotheses and predictions

- Hypothesis: A guess, or conjecture, or provisional statement about how/why you think something works
- Prediction: If your hypothesis is true, then what would the results look like? What data would support the hypothesis?

Hypotheses and predictions

- Hypothesis should describe a general pattern or relationship and provide a reason why you expect this pattern
- Prediction should explain what observation or result would support this hypothesis

Example hypothesis and prediction

- Hypothesis:
 - Male bullfrogs call more frequently when rivals are present because increased calling improves their chances of attracting females
- Prediction:
 - In experimental enclosures, male bullfrogs will call more frequently when playback of a rival male's call is broadcast compared to silence.



Hypotheses and predictions

- Hypothesis(es) should:
 - Describe a pattern or relationship that you expect and reason why
 - Be testable - a study can be designed to assess evidence for hypothesis
 - Be specific - not just a general hypothesis but one that applies to your study system(s)

Hypotheses and predictions

- Predictions should:
 - Explain what results you expect to happen if your hypothesis is supported
 - Describe your behavior(s) of interest in quantifiable terms
 - I.e., not just “social behavior” but “time spent within one body length
 - Not just “foraging behavior” but “number of food items consumed and diversity of species consumed”

Hypotheses and predictions

- Due next Friday, **October 10th** at 11:59 pm
- Worth 40 points (10% of proposal grade)
 - Can build on existing ideas for hypotheses from first two proposal assignments

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